

Final M425 (I)

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1 Introduction

Diffusion-weighted imaging (DWI) is a magnetic resonance imaging technique that is used to study the structure of white matter in the brain. However, methods such as diffusion tensor imaging (DTI) are limited—they only use a few gradient directions, which fails to capture complex fiber structures. To improve this, high-angular-resolution diffusion imaging (HARDI) methods have been developed, but they require long scan times.

This paper proposes a technique to enhance the angular resolution without additional scanning. By analyzing patterns in the raw HARDI signal, the method predicts new diffusion directions and synthesizes the corresponding signals using local signal similarity. These synthesized signals are then used to compute orientation distribution functions (ODFs) with improved accuracy.

2 Theory and Method

The core idea is to use known signals in measured diffusion gradient (DG) directions to estimate signals in unmeasured directions. These predictions rely on the idea that, within a local region of space, diffusion behaves similarly along neighboring directions.

For each unmeasured target direction, we then select a set of measured neighboring directions from the DG direction set $\mathcal{D}_N$. From this, triangles such as $\mathcal{N}_{\mathbf{v}_j}^{3,1} \subset \mathcal{D}_N$ are formed to define local geometric neighborhoods. The reliability of a triangle is assessed using correlation coefficients computed across a $3 \times 3 \times 3$ local window. These coefficients evaluate the similarity between the local diffusion patterns in the directions under consideration. If all pairwise correlations within a triangle are positive, it is deemed suitable for interpolation.

A new appropriate direction $\mathbf{u}_c$ is then determined as a weighted combination of the triangle's directions, lying within their convex hull:

$$\mathbf{u}_c = \sum_{k=1}^K w_k \mathbf{g}_{jk}, \quad \text{where } \sum_{k=1}^K w_k = 1, \quad w_k \geq 0$$

Once the direction is predicted, the signal at $\mathbf{u}_c$ is synthesized by taking a corresponding weighted combination of the signals in the neighboring directions:

$$S(\mathbf{u}_c) = \sum_{k=1}^K w_k S(\mathbf{g}_{jk})$$

To extend the coverage of the signal over a finer angular grid, we then apply RBF interpolation. A Gaussian kernel is used to interpolate the signal at new directions using both measured and synthesized directions:

$$S(\mathbf{v}_j) = \sum_{g \in D_C} c_g \cdot \exp(-\alpha \|\mathbf{v}_j - g\|^2)$$

The interpolation weights $\mathbf{c}_g$ are solved from the linear system:

$$(A + \lambda I)\mathbf{c} = \mathbf{b}$$

3 Validation

To evaluate the effectiveness of the proposed interpolation method, the authors compare reconstructed spin distribution functions (SDFs) derived from three datasets:

1. f_{64} : Full measured signal (64 directions)
2. f_{32} : Partial measured signal (32 directions only)
3. f_{cm} : Partial signal + interpolated directions (32 + synthesized)

3.1 Relative L^2 Error

This is the primary quantitative metric:

$$RE_{32} = \frac{\|\tilde{f}_{64} - \tilde{f}_{32}\|}{\|\tilde{f}_{64}\|}, \quad RE_{cm} = \frac{\|\tilde{f}_{64} - \tilde{f}_{cm}\|}{\|\tilde{f}_{64}\|}$$

This measures how close the estimated SDFs are to the true distribution.

Crossing Fiber Voxels

The authors also calculate the percentage of crossing fibers that were detected in the brain from each datasets. Interpolated signals (f_{cm}) yield a higher percentage of accurately identified crossings than partial signals alone.

Angular Deviation and Fiber Ratio

The angular deviation between the estimated and true secondary fiber directions is calculated to assess directional fidelity. In Table 3 of the paper, the ratio of second fiber directions F_{d2} is reported for both f_{cm} and f_{32} relative to f_{64} , showing improved structural alignment using interpolated data.

Qualitative Comparison

Figure 4 of the paper displays voxel-wise visualizations of SDFs and their relative errors. These figures support the quantitative findings by highlighting spatial consistency improvements across the various data.